

Leveraging passive RFID technology for construction resource field mobility and status monitoring in a high-rise renovation project

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ABSTRACT

The hypothesis is that leveraging automated data collection technology for site status analysis would play a more significant role in advancing decision making in construction projects if applied to traditional labor intensive management work tasks. Such manual data record keeping is, for example, progress tracking measurements and reporting of daily work status and process flows. Recent research on material tracking has demonstrated that the implementation of automated material tracking technology is feasible. Studies have yet to demonstrate whether the same or other technology can be used on other resource types, including workers, and furthermore, in advancing technology that works bi-directional: (1) collect and analyze data, and (2) return automated feedback to the decision makers at the management or even the workforce level. Despite a rigorous cost–benefit, hardware reliability and safety tests, implementation of technology in field operations is often performed on an as-needs basis. Project based case studies are effective research tools to measure the benefits and barriers that technology comes with. This paper will demonstrate results to the design, development, and furthermore and mainly, the effective and affordable implementation of a state-of-the-art wireless passive RFID-based technology system that collects and distributes information from and to decision makers. The developed technology was tested for several consecutive months on more than 50 construction workers, material carts and work related items, and personnel and material lifts that were critical in a high-rise building renovation project. Metrics to measure success in the phases of data collection, the signal and data processing, and in the use of newly generated or already available information for advanced decision making based on passive RFID technology will be presented.

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1. Introduction

The analysis of labor productivity has mostly been done manually. Studies have shown that the manual analysis is time consuming, subjective to judgments and error prone [1–3]; therefore, there have been attempts to automate the analysis such as mobile tracking in lay down yards. Examples of these attempts include pre-cast concrete, fabricated steel elements, and automated tool tracking. Past research has also shown the multitude of benefits obtained from implementing RFID technology within various construction sites including asset and material tracking, inventory management, and on-site security upgrades [4–6]. Projects with higher automation and integration of information technology improved between 31% and 45% in productivity [7]. In material tracking studies, the use of RFID allowed laborers to locate materials with an improvement ratio of 8:1 over manually tracking. RFID implementation in an outdoor environment led to a 4.2% increment in steel erection productivity [5]. The technology started getting more complex and began

tracking materials, workers, and equipment in real-time. Such technology includes ultra wideband [8,9], GPS [5], and multiple sensors [10]. However, such research has only been utilized in horizontal construction, but not in the vertical construction types.

Additional construction improvements can be identified in terms of productivity analysis of work crews, material transport, and the overall approach to a project in order to determine whether the construction process is operating at maximum efficiency or can be adjusted to improve its effectiveness. In particular, status of labor forces is an important area to maintain for completing a project on schedule. A large percentage of construction costs result from the quantity of labor hours spent completing the desired tasks. Thus, by maximizing labor productivity, companies can avoid additional costs due to falling behind schedule. Many factors determine whether workers are able to complete their tasks at the necessary pace including experience, age, skill, motivation, and leadership. Appropriate work conditions can also influence the entire operation, such as project size and complexity, site accessibility, labor availability, equipment use, and local climate [11,12].

This paper presents the objectives of the research and the results to the tests of our hypothesis, which was to see whether or not the automated RFID data collection technology could provide important

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information to improving resource status and mobility. The implementation of low-cost, passive radio frequency identification (RFID) technology and a study of labor status analysis for a window replacement project on a high-rise construction site are also included in this paper. This study provides detailed studies to track the efficiency of a buck hoist operator and material lift system for transportation and illustrates the applicability of the technology despite the presence of numerous signal impeding obstacles located throughout the site. These issues are resolved with an effective automated location and time tracking system that works in both an indoor and outdoor environment simultaneously with a data recording software and database. Both will be presented and relevant information regarding system optimization and worker feedback for future use will be recognized. The contributions of the research will include the results of the testing, an assessment of the RFID system, and recommendation for future applications and studies.

2. Background

2.1. Lifts and hoists in high-rise construction

Thousands of tower cranes are currently employed around the world and are generally regarded as a major item of equipment on high-rise building sites. While tower cranes possess large horizontal and vertical lifting capacities, operational inefficiencies and miscommunication cause time delays, resulting in scheduling and safety issues [13–16]. Efficient planning and preparation is vital to maximizing productivity, as the cycle time for a single lift can surpass 20 min in a building consisting of 40 stories. To mitigate some of the inefficiencies, Lee et al. [17] developed a tower crane operating system that uses wireless video control and RFID technology to provide real-time visual images, details of materials waiting to be lifted, and the locations for installation. Based on a case study, benefits that resulted included increased work speed, greater communication, and improved safety. Robotic tower cranes have also been proposed ([18,19]) and, at some point in the future, are expected to have key advantages and improvements in construction scheduling, labor and material costs, and safety. By using a series of laser devices, GPS, encoder, accelerometer, and other devices, improvements in productivity between 10% and 50% are expected. However, limited availability and poor weather conditions may make this type of crane technology inoperable, and thus irrelevant in many regions of the world. Buck hoist operations in high-rise construction, in particular in renovating existing buildings, provide an alternative to the use of tower cranes.

2.2. RFID technology in construction

Radio Frequency Identification (RFID) is communication via radio waves (see example in Fig. 1). RFID systems, classified as active (battery powered), semi-passive (battery-assisted), or passive (without battery), are composed of a transponder and transceiver (reader) that gather and transmit information wirelessly to a RFID tag, often without the need of a direct line-of-sight to the tags. Each RFID tag can have its unique identification number. Ross et al. [20] tested passive ultrahigh frequency tags for durability of RFID tags in various harsh conditions and found the tags were durable enough to work

despite the existence of extreme moisture, pH, temperature, and pressure. While bar code systems are limited due to line-of-sight, durability, and read-range constraints, RFID technology provides significantly greater read-ranges and works under rugged outdoor and indoor conditions including in temperatures from -40°C to 200°C [20].

Active RFID has been previously tested for outdoor construction. Active RFID technology can simultaneously and uniquely recognize facility items, store information regarding maintenance history of these items, and continuously update the information in real-time [21,5].

Passive RFID tags are most suitable for an indoor construction application because of various benefits: (a) reduced cost (less than \$0.20 for a typical tag), (b) small in size (flat like a sticker), (c) extended operational life (no battery), and readable and writable data storage capacity of roughly 128 to 256 bytes (which can be hyper-linked to other information or a database). The biggest drawback of passive RFID tags is that they have no self-reporting capability. The read ranges of passive tags for construction applications reaches currently around four to ten meters (depending mostly on antenna type). However, performance is reduced in environments that cause multipath or those in close proximity to any metal surface as the metal causes signal attenuation [22].

2.3. Case study

The site of the experiment in downtown Atlanta served as an excellent case study as it involved a 73 story structure composed of concrete and metal, located in a densely populated city. The Westin Peachtree Plaza Hotel, located in Atlanta, Georgia, is the second tallest all-hotel skyscraper in the western hemisphere. In 2008, a category EF2 tornado made a direct hit on the hotel, destroying over 500 windows (see Fig. 2). Skanska USA Building Inc. was the main contractor, and the decision to replace all 6350 windows (new building and energy efficiency codes) was enacted. Buck hoists were used as the primary transportation of resources. The design of the hotel, with its protruding skylight, made the possibility of having one complete track impossible. Therefore, a platform was created above the skylight on the 16th floor for the transition of resources from the street to the upper floors. There were two hoists that led from the street to the platform, and two that led from the platform to the upper floors.

3. Research scope, objectives, and methodology

The research scope focused on a high-rise renovation project, which is centered on vertical construction. The general contractor had to replace 6350 windows within a 15 month project duration. A lack of open space at the base of the high-rise made the use of cranes unreasonable because the outriggers would have extended into the street, which would block the heavy traffic. In order to meet the needs for the glass replacement, one set of two buck hoists had been placed at the street elevation which then rose to a transition platform on the 16th floor. Then a second set two of buck hoists went from the platform up to the working floors. The payload capacity of the hoists was over 4000 lb and provided transportation for large materials with complete accessibility to each floor of the building. Height (or length) of a building in construction largely affects the



Fig. 1. Wireless radio frequency identification (RFID) system.

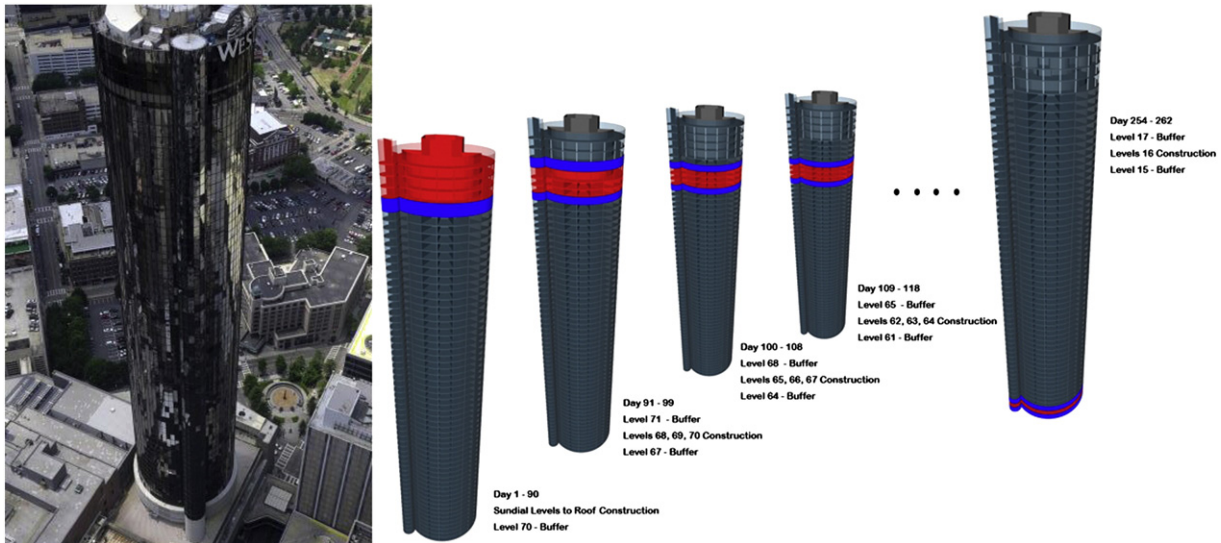


Fig. 2. Work sequence and buffer zones in a high-rise renovation project (image courtesy: Skanska USA Building Inc.).

efficiency of lifting (or moving) equipment installed on-site, thus affecting the overall project schedule [12].

The goal of this research was to test the viability of leveraging passive RFID technology installed within a construction site, and to see whether or not the important information drawn from the research can aid in improving resource status and mobility, in addition to help insure that time wasted during construction is minimized. Time wasted, or rather 'lost time', is defined as any amount of time not being used in the performance of a task for the project, negatively affecting productivity. Past research has shown that RFID technology is valuable in material and tool tracking as well as eliminating time spent with manual data entry. In this research, each of the construction workers, materials, and elevator buck hoists on-site were tagged with passive RFID. The objective was to analyze the efficiency of using a buck hoist system installed on the exterior of a building. The vision behind this research was to reduce human error, measure personnel check-in and check-out time, measure buck hoist operation time and availability, and to control site access and security.

A commercially-existing RFID hardware technology was selected, but since unavailable, data gathering software and user interfaces optimized for a high-rise renovation project were developed (in Visual Basic) to capture the RFID tag reads. These reads were then linked to a database that recorded the exact read times for the worker, buck hoist, and material movement throughout the high-rise. The scope of the software technology was designed to illustrate the efficiency of the buck hoist system rather than demonstrating inventory, transport of materials, or personnel productivity. Although direct measure of exact location of material or productivity of personnel is not feasible with existing passive RFID technology (unless a complex system of readers is installed on working floors), the data gathered allows to infer probable solutions to existing problems workers and decision makers face every day, e.g. waiting for buck hoists to arrive, where-about of workers, etc.

The research focused on the automated data gathering and analysis of project-critical resources in a high-rise renovation project. The project type, by itself, made this research to be a first of its kind. Worker helmets, lifts, material transport carts, and windows were tagged with unique passive RFID tags and tracked in real-time by strategically placed RFID readers (see Fig. 3). By placing RFID tags on each buck hoist, time studies could be conducted to determine the amount of time it took to travel, as well as time spent waiting for lifts to arrive. Various factors affecting the construction process and whether RFID technology has the capacity to improve resource mobility and status monitoring despite the presence of signal inhibiting materials.

The use of RFID will exhibit the benefits and drawbacks of the buck hoist material and personnel lift system in a high-rise construction setting. The implementation of RFID technology on buck hoists, personnel hard hats, and material and material carts will determine whether a high-rise renovation project can support and benefit from the implementation of such technology as well as the decision when to use a buck hoist for personnel or material transport. The ability of the RFID technology in this work environment will be shown after implementing the necessary means for acquiring sufficient data, conducting analysis of the data, and identifying the problem or areas with room for improvement.

To minimize capital investment, wireless passive RFID technology was implemented only on each working and transition floor. Four wireless RFID readers were installed on the platform facing each door so that all the entry ways were monitored (see Fig. 3b). One wireless RFID reader (with extension antenna) was installed inside the first room on each of the three working floors (see Fig. 3d). As the working floors moved overtime, these three readers redistributed on each working floor. Additional antennas were installed in the same manner on the three working floors so that all entry ways to and from the buck hoist were monitored to track the in- and out-going resources.

Hoist three was designated as the main hoist used for the transportation of resources from the platform to the upper floors. To track the movement, eight tags were placed on the buck hoist, four on each door. Fig. 3i shows the placement of the tags, which were strategically placed so that the tag on the hoist door would still be read by the reader even if the door was opened.

Workers were tracked by placing two tags on the helmets (see Fig. 3e and f). The helmets were chosen as the means for the placement of the tags because a worker always wears it at the job site. Each tag on a helmet is unique to a worker and is often personalized with work stickers and decals (some workers were proud to display the unique design of the passive tag). The use of multiple stickers per single item was to guarantee that it was able to be picked up by the readers from all angles. Carts and windows were also tagged with multiple tags during the preliminary field testing of the technology and to ensure readability of the tags (see Fig. 3g).

4. Field trial and results

4.1. Details to implementation of hardware and software technology

The first step in implementing the developed wireless passive RFID tracking system was to issue each worker, material cart,

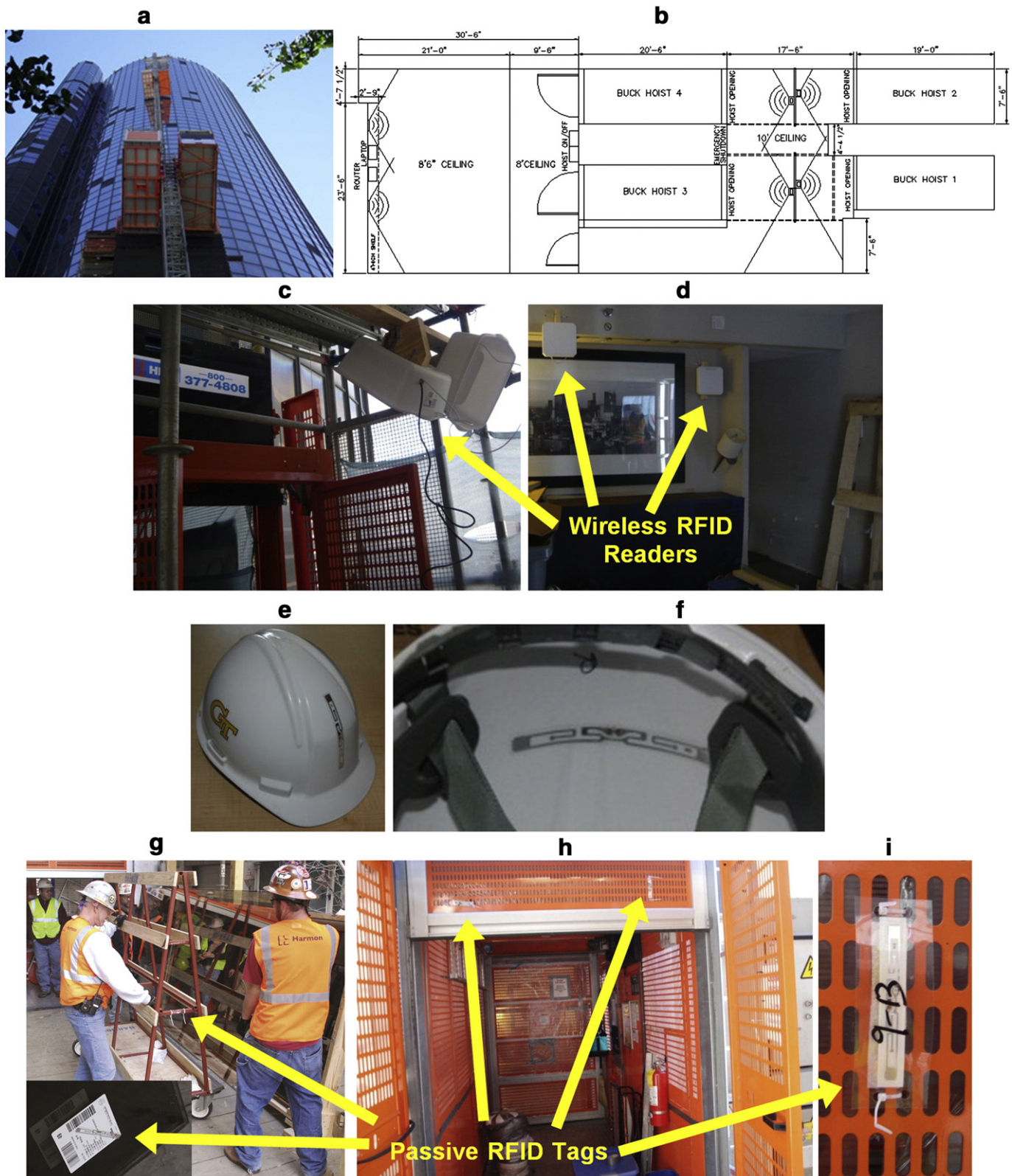


Fig. 3. Installation of RFID system: (a) two pairs of two buck hoists reached from the ground to the 16th floor, and from the 16th to the upper floors; (b) plan view of buck hoists area and entry room into building; wireless Gen 2 RFID readers were strategically placed on the 16th floor at a secure, off-the street location and inside the first room of each working floor; (c) wireless RFID reader in front of the entrance to a buck hoist; (d) wireless RFID reader in the entry room to each floor; (e) passive RFID tag attached to the outside of a construction helmet; (f) passive RFID tag attached to the inside of a construction helmet; (g) passive RFID tag attached to a material (window) and a material cart; and (h and i) passive RFID tags attached to the entrance door of one of the buck hoists.

replacement window, and buck hoist one or multiple unique RFID tag(s). Each construction worker was issued two unique RFID tags for redundancy. One RFID tag was placed on the outside the

front of the helmet while the other tag was placed inside the rear of the helmet. Each tag was given a unique ID number in order to help maximize detection and to allow data collection to tags that could

not be read. In addition, signal strength between the two tag locations on helmets was measured to identify the best possible location of passive RFID tags on helmets. Additionally, each buck hoist had eight RFID tags attached to its entry doors. Four tags were placed on each of the front and back entrances to maximize detection in a metal platform environment. These tags occupy each corner of the hoist door facing the incoming radio frequency signals from the antennas. Glass that was destined for replacement arrived at the construction site in large wood-framed batches, and then was separated into groups of six. Each batch was tagged with a single unique RFID tag.

The issue of privacy with tracking workers (human subjects) had been resolved by strictly using volunteers in accordance with rules set by the universities internal review board. The workers were given an explanation of the research and the purpose and use of the tracking technology, along with the right of nonparticipation. The names given to those who did volunteer were aliases, and the information and whereabouts were for research use only and would not result in any incrimination.

The second step of the implementation was the strategic placement of six RFID readers (transponders), two mobile laptop computers, and two wireless routers to connect the RFID system. The glass replacement process and work zone consisted of five floors at a time, in which the middle three had access to the buck hoists. Actual work took place on the middle of three floors while the outer two floors were used primarily as buffer zones to control construction noise that may affect guests in the high-rise hotel. One reader was placed on each of the three work floors and faced the entrance of the buck hoist, and one laptop and router was placed in the middle of the working floor. Each antenna reader had a particular IP address that was used for identification, and allowed the database to display locations of the RFID tags when they appeared in the work floor. The antenna readers were connected wirelessly via a Wireless Local Area Network (WLAN) through an onsite router, then to the laptop computer that was used to store the collected data. The wireless routers on site had enough signal strength to penetrate five concrete slab floors.

The same connection methodology is implemented on the platform (16th floor) of the high rise building. The primary reason for deploying two separate systems is because it was tested impossible for a single router to broadcast a signal from the 16th floor to the top floors of the building; with a maximum separation of fifty floors. Two readers were located on the platform of the 16th floor with auxiliary add-on antenna extensions. These add-ons were used to increase the range and visibility of each of the readers that they were connected to. The extended add-ons also allow the antenna readers to have a field-of-vision of 120°, rather than the limited 60 degree scope of a single reader.

The third step to set up the wireless system was to successfully connect the antenna readers with the laptop computers to collect the data. Each laptop computer utilized software that was developed by the authors. The software allows the acquisition of data that consists of: (a) RFID tag ID, (b) read frequencies, (c) time of read instance, (d) signal strength and link quality indicator, and (e) antenna reader internet protocol (IP) address. Recorded data was stored as either an undesignated file type or text file format. Readings from the antenna readers occurred at the default read rate of the readers is set to 100 Hz and was set to record during regular work times (6 A.M. to 8 P.M. on any day to compensate for overtime and weekend work).

A user interface was created in order to make communication with the readers more easily accessible for the project manager. Although some commercially available RFID software exists, it does not address tasks or questions a construction engineer or manager would like to see resolved. Since it was unfeasible for this research project to hire a system integrator because of high cost, a hardware-

software interface was developed specifically to tailor the needs of the project manager from quick and easy setup of equipment to storing and analyzing valuable data. This user interface prototype was kept simple, but through computer programming, additional features can be added to suit a project manager's needs, such as automatic scheduling, visual location of site resources, and graphing capabilities. Fig. 4 displays a screen shot of the user interface showing its simplicity to a complex RFID system. The project manager can easily connect a reader, save files to a data base, and see what tags are being picked up with a click of a button.

The user interface allowed the project engineer to specify what data should be saved and where to save it to. The manager can choose specific data and retrieve results with any relevant information. This data was not in a workable form for the project manager and was often jumbled and stored in large files. Therefore, a database was also developed to store and organize the data into a more workable form, as well as making the information that has been tracked by the RFID readers easily accessible to a project manager. The database used a series of relationships and queries to link the tag data to other stored information within the database, such as worker and floor information. The developed interface and database were the pathways that enable the project engineer to access valuable information, saving both time and money. For this research, the database was created to store information regarding the movement of workers, lifts or hoists, carts, and glass. Fig. 5 illustrates a sample data set that becomes available when linking recorded information of the passive RFID system.

Data filtering was another important application that the database provided. The reader read and stored data up to a rate about 5 Hz, creating a large data set in short amount of time that took up a lot of space on the computer hard drive. There have been a few instances in which a data overload caused the program to crash. Data filtering was important because it drastically reduced the large data file by eliminating redundant information. For example, a tag would be read about a hundred times before it moves out of the "line-of-site", and having multiple tags read simultaneously only added to the huge amount of data. The solution was to apply empirical thresholds to only keep the first and final reads, eliminating the redundant data in between (signifying the entrance and exit times). That way, algorithms would show that a tag is there without the large amounts of data. Further use of signal strength to detect the direction of the tag (resource) movement was not conducted (although it is technically possible, see Fig. 5). The possibility of the actual movement of the tag, with respect to only one RFID reader is possible. Although the reader did present the signal strength that has been received after bouncing back from the tag, there were many factors that were not resolved in this study and prohibited recording the actual direction of a tag (e.g. metal interference, field of tag view for the RF signals, etc.). The possibility to pinpoint the location and direction of a tag requires more sophisticated hardware and set up, such as potentially multiple readers for triangulation.

Data filtering also eliminated random and impossible reads, which were referred as "read phenomena." For reasons yet to be determined there have been several occurrences where tags would appear on multiple floors simultaneous, or would jump floors in an extremely short amount of time. For instance, a buck hoist would be on floor 62, and within those group of reads that verify the position, one or two reads would show it appear on a different floor for a split second. The threshold of the random reads was less than 1% of the total, accurate reads. Speculations of the phenomena include interference from the steel scaffolding or wireless network issues. An algorithm was developed and used to verify the position of the buck hoist and to eliminate the random reads. The algorithm would scan the read data and filter out any misplaced data points. The basic concept was to compare a group of sequential data points, and if the first two adjoining reads were different floors, then it looked to the third and fourth

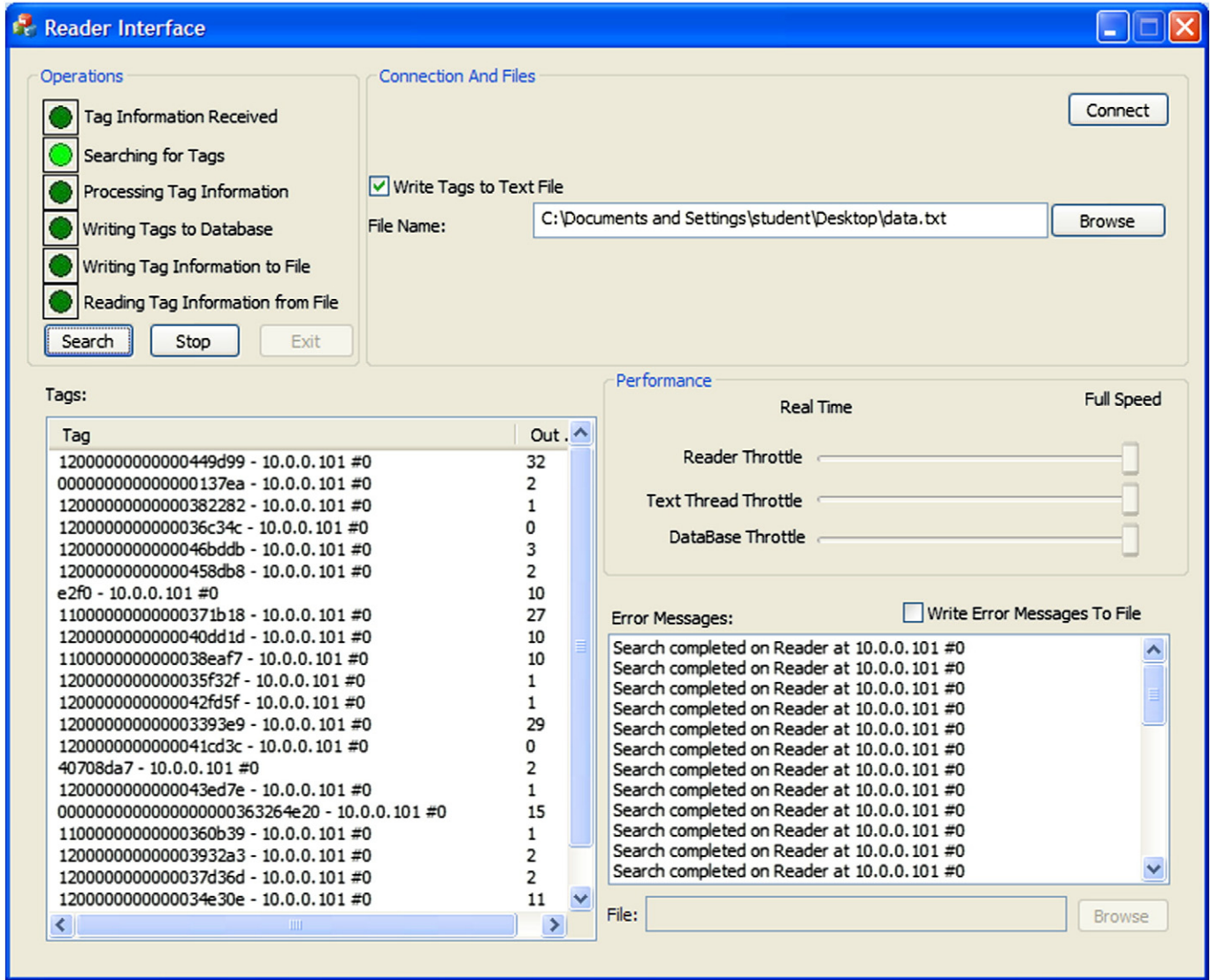


Fig. 4. Screen shot of the user interface to connect and read data from a RFID system.

reads to verify if the floors have changed or not. If however, the third and fourth read were not the same as the second then it eliminated the random read. Another example was that if two adjoining reads were differentiated by less than a set time (depending on the

speed the hoist travels between different floors), the algorithm would look a few reads before and after to verify the position of the buck hoist, and then delete the random read, since the travel time was less than the speed of the buck hoist.

Worker ID	Last Name	First Name	Tag ID	Timestamp	Reader ID	Description	Signal Strength
4	Doe	John	10000000000000045afb	02:04:18	192.168.1.52 #0	Floor 64	70
4	Doe	John	10000000000000045afb	02:07:00	192.168.1.52 #2	Lift 1 Platform	85
3	Doe	John	10000000000000032a1c	02:07:01	192.168.1.52 #2	Lift 1 Platform	97
3	Doe	John	10000000000000032a1c	02:07:01	192.168.1.52 #2	Lift 1 Platform	77
4	Doe	John	10000000000000045afb	02:07:05	192.168.1.52 #2	Lift 1 Platform	73
4	Doe	John	10000000000000045afb	02:07:05	192.168.1.52 #2	Lift 1 Platform	78
3	Doe	John	10000000000000032a1c	02:07:05	192.168.1.52 #2	Lift 1 Platform	72
3	Doe	John	10000000000000032a1c	02:07:06	192.168.1.52 #2	Lift 1 Platform	80
3	Doe	John	10000000000000032a1c	02:07:13	192.168.1.52 #2	Lift 1 Platform	75
4	Doe	John	10000000000000045afb	02:07:13	192.168.1.52 #2	Lift 1 Platform	71
4	Doe	John	10000000000000045afb	02:41:10	192.168.1.52 #0	16th Floor Indoor	71
3	Doe	John	10000000000000032a1c	02:41:11	192.168.1.52 #0	16th Floor Indoor	72
3	Doe	John	10000000000000032a1c	02:41:13	192.168.1.52 #0	16th Floor Indoor	74
3	Doe	John	10000000000000032a1c	02:41:13	192.168.1.52 #0	16th Floor Indoor	83
4	Doe	John	10000000000000045afb	02:41:13	192.168.1.52 #0	16th Floor Indoor	73
4	Doe	John	10000000000000045afb	02:41:15	192.168.1.52 #0	16th Floor Indoor	69

Fig. 5. Sample data of two corresponding tags to one worker (worker ID, name (anonymized), passive RFID tag ID, timestamp [hh:mm:ss], RFID reader ID, reader description, signal strength of reading).

4.2. Results

The value of producing a database was evident from the ability to select specific data from a variety of options for data analysis. The best practice was to give a consultant or staff member adequate time to find valuable resource mobility or status data and prepare the necessary steps to implement changes for improvement. Automating the analysis of recorded data saved significant time when examining labor productivity. In a high-rise construction setting consisting of workers, manually performing time studies was an inefficient and lengthy task. Thus the need for an automated data analysis tool was necessary to produce valuable time study results in a timely manner.

Depending on the desired information, supervisors or managers could conveniently edit this table to store additional information. Other relevant information, such as weather conditions, could be linked to queries by date to assess if those weather conditions factored into the results obtained by the readers. Additionally, each reader had been programmed to record the timestamps for the exact moment in which the tag was read.

4.2.1. Tracking workers

The example in Fig. 5 shows that the time it took a worker (or any other resource the tag is attached to) to use a buck hoist can be recorded. The figure displays the results to consecutive tag readings of a worker that left from the 64th floor at 2:04:18 P.M. and arrived at the 16th Floor Platform at 2:07:00 P.M. These two timestamps can be used to find the travel time for this distance which is calculated to be 2 min 42 s. Since tags were continuously being recorded when in close proximity to a reader, it is possible to use this analysis tool to find additional information such as the time lost when a worker was waiting for a lift to arrive. In this instance the worker was walking inside the building once the worker arrived on the platform level.

Fig. 6 depicts another scenario that can be utilized for field mobility measurements. In this case, the worker passed the 16th floor at 01:57:32 P.M. and waited on the platform for 1 min 17 s before leaving the platform at 01:59:00 P.M. The worker then arrived at his desired location, Floor 58, at 02:04:00 P.M. which calculates that this hoist took the worker 7 min 17 s to reach the desired location. This tool allows travel time analysis to be conducted without the need for positioning multiple people throughout the site to record timestamps of specific workers or issues related to manual data entry errors.

4.2.2. Tracking equipment

Manual observations made it possible to document applicable data about which buck hoists were utilized most often. Through the implementation of the previously described RFID data collection system, the location of the four buck hoists on site were tracked and documented. Utilizing the developed RFID system made it possible to collect and analyze raw data automatically (raw meaning that no algorithms were in place to filter out any errors or redundant data).

Buck hoist three was the primary method of transportation from the 16th floor, the main staging platform, to the upper three working floors. Over a typical working day, the buck hoist made several trips to the upper floors to bring or pick-up workers, equipment, and/or materials to their working destination. The movements of the buck hoists were recorded on a continuous work day basis, even on weekends that had no working action. With longer spans of data, such as a weeks or months, more observations were performed and allowed for more accurate conclusions. Fig. 7 shows the travel patterns of buck hoist three over one complete (sample) working day. Floor sixteen was the start point of the buck hoist while floors 48 to 50 were the working floors during this day. Based on the data analysis that were automatically synchronize the passive RFID tags of workers, equipment or materials, cycle times and usage of the buck hoist can be calculated. Data to wait times, peak times, and underutilized times can be recorded and utilized.

A particular important piece of information for project managers that resulted from the data analysis was the cycle time of the buck hoist, including its cycle times per day, week, month, and even over the entire project. For example, on April 15, the average cycle time of a worker that went from the 16th to the 49th or 50th floor, then loaded workers or materials, and then returned back to the 16th floor was 8 min 24 s. Compared to other days, the average amount of time the buck hoist was in use on a typical working day during the week of April 26 to April 30 was 2 min 30 s. Such information can play a vital role in decision making to save time and money, including helping construction managers to allocate resources, keep project schedules on time, and improve efficiency.

Knowing the amount of time spent idle and at what floor of the project can help with more effective planning and project management. The developed data analysis system allowed the measurement of the time that a buck hoist was in idle position. Over a typical work week, buck hoist three spent 1 min and 30 s idle at the 16th floor. The 16th floor was the default location for the buck hoists to remain in idle position. Additionally, the buck hoists sometimes stayed idle on upper

Worker ID ▾	Last Name ▾	First Name ▾	Tag ID ▾	Timestamp ▾	Reader ID ▾	Description ▾	Signal Strength ▾
3	Doe	John	1000000000000032a1c	01:57:32	192.168.1.50 #0	16th Floor Indoor	74
4	Doe	John	10000000000000045afb	01:57:43	192.168.1.50 #2	Lift 1 Platform	68
3	Doe	John	10000000000000032a1c	01:57:45	192.168.1.50 #2	Lift 1 Platform	66
3	Doe	John	10000000000000032a1c	01:57:53	192.168.1.50 #2	Lift 1 Platform	76
4	Doe	John	10000000000000045afb	01:57:54	192.168.1.50 #2	Lift 1 Platform	77
4	Doe	John	10000000000000045afb	01:57:57	192.168.1.50 #2	Lift 1 Platform	77
3	Doe	John	10000000000000032a1c	01:57:57	192.168.1.50 #2	Lift 1 Platform	70
3	Doe	John	10000000000000032a1c	01:58:07	192.168.1.50 #2	Lift 1 Platform	71
4	Doe	John	10000000000000045afb	01:58:08	192.168.1.50 #2	Lift 1 Platform	77
3	Doe	John	10000000000000032a1c	01:58:12	192.168.1.50 #2	Lift 1 Platform	77
4	Doe	John	10000000000000045afb	01:58:12	192.168.1.50 #2	Lift 1 Platform	75
3	Doe	John	10000000000000045afb	01:58:19	192.168.1.50 #2	Lift 1 Platform	75
4	Doe	John	10000000000000032a1c	01:58:19	192.168.1.50 #2	Lift 1 Platform	72
3	Doe	John	10000000000000045afb	01:58:22	192.168.1.50 #2	Lift 1 Platform	73
4	Doe	John	10000000000000032a1c	01:58:22	192.168.1.50 #2	Lift 1 Platform	74
4	Doe	John	10000000000000045afb	01:58:22	192.168.1.50 #2	Lift 1 Platform	73
3	Doe	John	10000000000000032a1c	01:58:40	192.168.1.50 #2	Lift 1 Platform	72
3	Doe	John	10000000000000032a1c	01:58:49	192.168.1.50 #2	Lift 1 Platform	72
3	Doe	John	10000000000000032a1c	01:58:59	192.168.1.50 #2	Lift 1 Platform	69
4	Doe	John	10000000000000045afb	01:59:00	192.168.1.50 #2	Lift 1 Platform	76
4	Doe	John	10000000000000045afb	02:04:00	192.168.1.50 #0	Floor 58	72

Fig. 6. Worker wait and travel time for elevator.

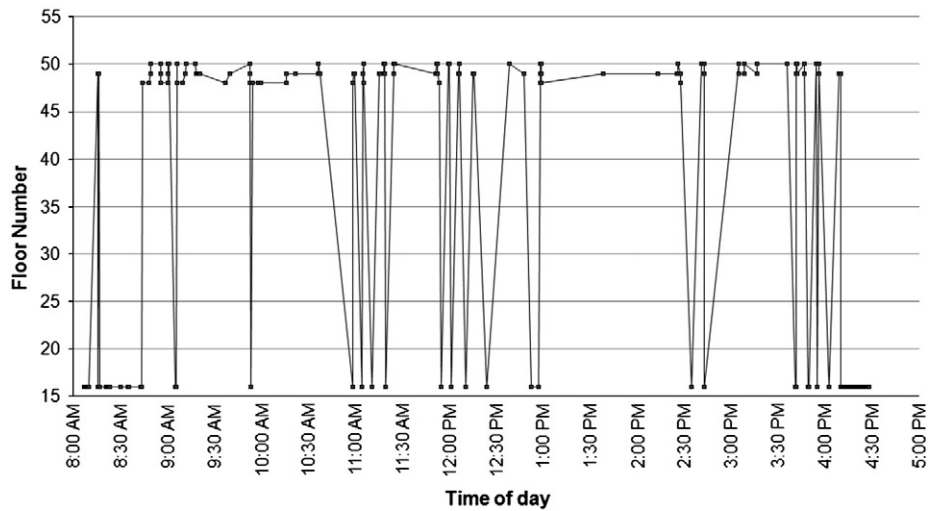


Fig. 7. Travel patterns of buck hoist three for one day.

floors. For example, during the week April 26 to April 30, buck hoist 3 spent on average 41 min 5 s in idle position on the upper floors.

On certain days of the week, the tower on which the buck hoist operates would experience servicing and maintenance downtime. During these times, the buck hoist remained primarily on the top working floors and rarely made any trips to the 16th floor. Fig. 8 shows the inactivity of the buck hoist, which was a day in which the top parts of the buck hoist tower were being disassembled.

On some very few working days on the critical path, workers and materials had to be corralled into a service elevator that was shared with hotel employees on the site. During these days, wait times for vertical transportation on the site significantly increased, leading to project delays and tension among the subcontracted workforce who might have been hindered in performing their tasks. Fig. 9 shows the average service elevator wait times during such sample days. In the pessimistic case, it took more than 16 min of wait time for the service elevator to arrive. The service elevators inside the building held less weight, and thus less workers and materials. With work tasks being on the critical path and thus highly dependent on each other, work progress was observed to slow down. Overtime and weekend work was performed to compensate for any delays and to take advantage of better than expected weather conditions. Efficiency can be maximized by increasing work time and reducing wait time. Fig. 9

shows the amount of time that was spent by the workers waiting to use the hoist. Graphs like the figure can be used by project engineers to see what delays exist, link to the causes, and take appropriate actions.

4.2.3. Tracking material

Glass was the major material that was moved among the site. Since the construction project was a glass removal and replacement job, the essential tasks revolved around the transportation of the glass when it arrived on site. A passive RFID tag was placed on the glass upon arrival at the street level of the job site.

The placement of the tag on the glass was important in itself, as the thickness of the glass prevented some RFID tags from being recognized by the readers. The sample of cycle times varied over different delivery dates, and was mostly influenced by the floor that it was hoisted to and the time that the glass delivery truck arrived. Cycle times did remain very similar to normal cycle times for the buck hoists, only adding a few minutes to account for loading and unloading of the glass carts. A graph would look similar to Fig. 6.

4.3. Assessment of passive RFID technology

One concern when implementing a RFID system on a construction site is ensuring the signals from passive RFID tags can be effectively

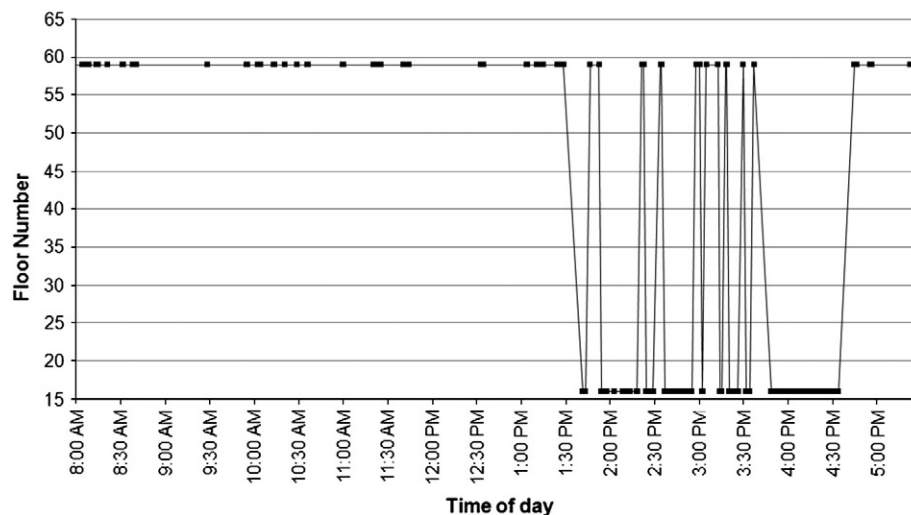


Fig. 8. Buck hoist remaining primarily on top level floors.

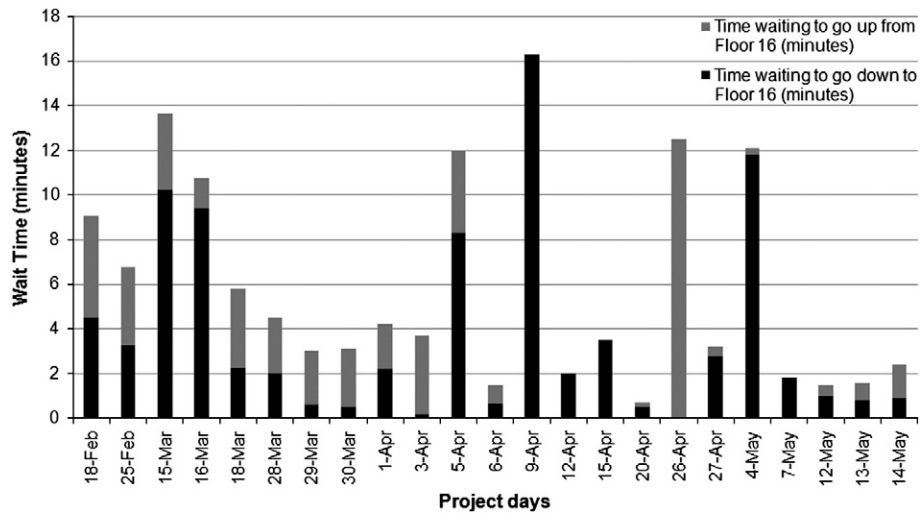


Fig. 9. Average worker wait time for service elevator.

transmitted and received. To overcome radio frequency signal interference issues, all four buck hoists were equipped with several passive RFID tags. Each hardhat of a worker was labeled with two inexpensive passive RFID tags. One of which was placed on the front of the hard hat, and the other was placed on the inside towards the back of the hard hat. With each passive RFID tag having their own unique ID, their readability can be analyzed and used as reference for future studies. Fig. 10 shows the location of two passive RFID tags mounted on front side and the rear inside of a hard hat. Since the passive RFID tag does not carry any battery, its adhesive and flexible label can very easily be attached to a hard hat. The readings in Fig. 10 show that the location of passive RFID tags at the front and rear of a helmet have little to no role in reading the passive RFID tag via the developed wireless passive RFID reader system. The two tags were found to be almost identical in the number of times read. The tag on the front of the helmet had 2872 reads. It was read a bit more often than the one on the back of the helmet with 2,745 reads. Redundant information from multiple passive RFID tags was preferred, but duplicate data can be easily filtered.

4.4. Automated field reports

4.4.1. Data visualization

An automated RFID system can help in eliminating the need for manual record keeping, or at least in assisting in this labor intensive, error-prone task. It allows a project manager or observer to see on which floor resources are located and to develop automated reports of past work days. Fig. 11 shows a visual for an automated report for a buck hoist and its related data. A total of 43 trips were made on the selected day. The average travel time of a trip of the buck hoist was 1 min and 34 s. The average travel velocity was 2.45 m/s. Such valuable information becomes available with the use of the system to maximize efficiency, which ultimately saves time and money.

The measured vs. planned utilization rate of hoisting equipment, in particular hoisting equipment, is often critical in resource allocation. Decisions in day-to-day operation eventually can be influenced once previously unrecorded data becomes available. A project manager, for example, could look back to a specific day and see why the hoist was not used or not available during the day of a material

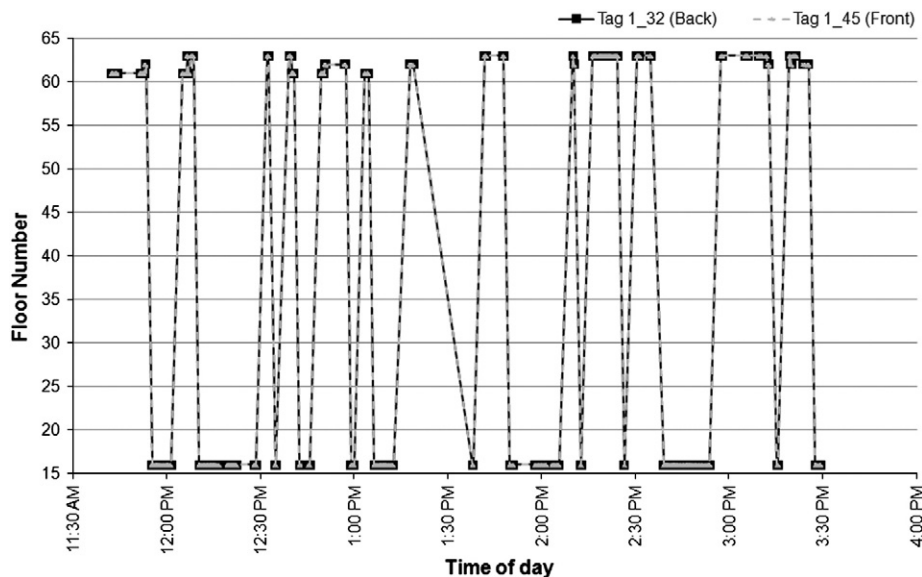


Fig. 10. Comparison between tags placed on front outside and back inside of the helmet.

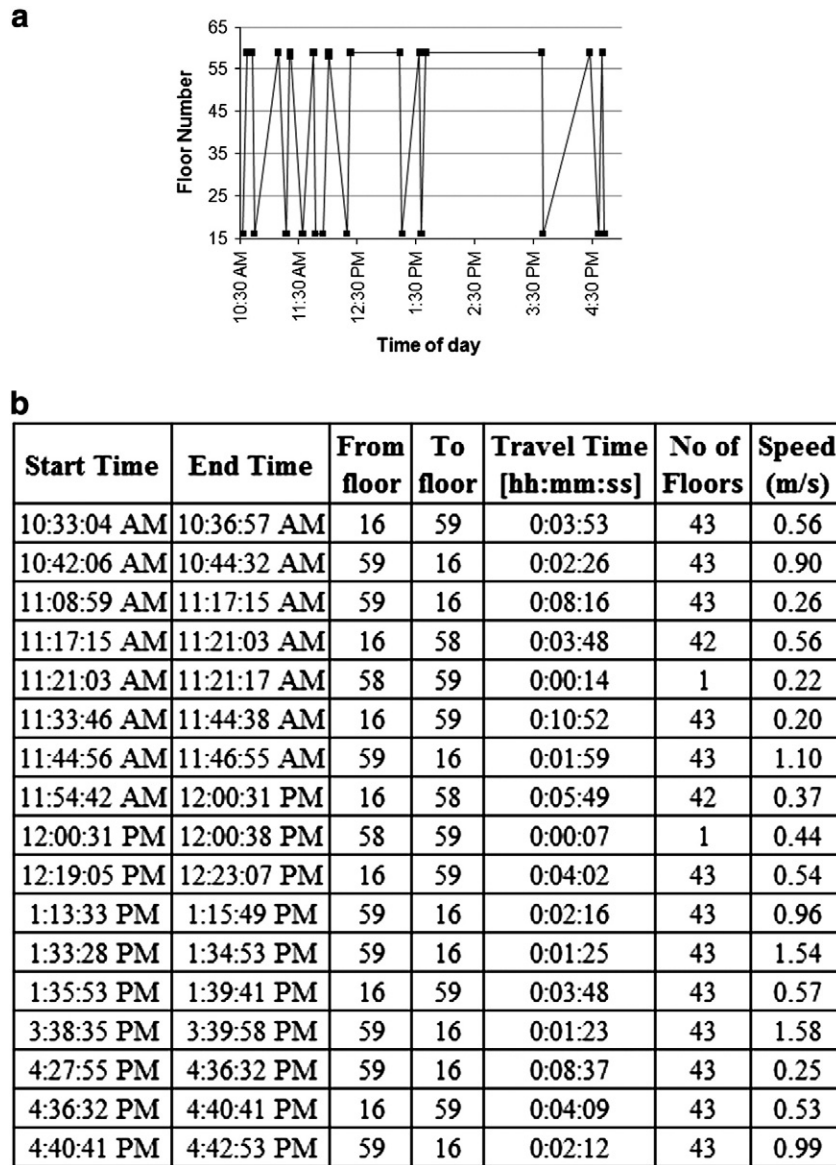


Fig. 11. Automated reports: (a) a visual and data to operation and (b) cycle frequency analysis of hoist three.

delivery to site, and what consequences it had on the overall operation of the site. Similar automated reports can be used to analyze the usage of resources over a longer duration of time. Trends can be determined for the current project, but information can also be used for scheduling the next project. The availability of hoists can thus be estimated (simulated) before a construction project starts, for example at the front-end planning stage, thus by maximizing efficiency avoid cost and delay.

4.4.2. Utilization rate of equipment

The travel statistics of hoist three for random days during the renovation project duration are shown in Table 1. The start time indicates when the buck hoist made its first trip and the end time relates to the time when the buck hoist made its last trip, respectively. The number of raw RFID data reads indicates the number of tags that were read using this tag (including duplicated reads). The error rate of the filtering algorithm that was applied (basic threshold) is presented. Typically, only a few data

Table 1
Analysis of trips made by hoist #3.

Date	Feb 22	Mar 9	Mar 16	Mar 30	Apr 7	Apr 23	May 18	Jun 8	Jun 15	Jun 17	Jun 28
Start time [hh:mm:ss]	11:42:24	07:18:46	09:28:53	07:45:09	08:07:41	08:06:52	08:14:55	08:18:10	08:00:01	10:50:18	08:25:19
End time [hh:mm:ss]	18:29:59	17:27:23	18:30:47	17:10:58	17:26:23	16:28:19	17:04:59	17:17:25	16:42:05	18:35:12	17:57:57
Work duration [hh:mm:ss]	06:47:35	10:08:37	09:01:54	09:25:49	09:18:42	08:21:27	08:50:04	08:59:15	08:42:04	07:44:54	09:32:38
Raw data reads [no.]	5141	766	501	1082	1001	595	576	2434	2389	2173	2541
Reads filtered [no.]	4	0	0	0	56	8	8	0	3	0	3
Reads filtered [%]	0.08	0.00	0.00	0.00	5.59	1.34	1.39	0.00	0.13	0.00	0.12
Trips made [no.]	50	16	17	92	120	55	61	18	28	9	22

Table 2
Daily utilization rate of the hoist #3.

Date	Feb 22	Mar 9	Mar 16	Mar 30	Apr 7	Apr 23	May 18	Jun 8	Jun 15	Jun 17	Jun 28
Loading and unloading time [hh:mm:ss]	00:00:30	00:00:30	00:00:30	00:00:30	00:00:30	00:00:30	00:00:30	00:00:30	00:00:30	00:00:30	00:00:30
Work time [hh:mm:ss]	01:40:40	00:57:01	01:13:46	02:28:35	04:10:16	02:16:18	01:13:46	00:47:28	01:39:41	00:29:47	00:52:51
Idle time [hh:mm:ss]	05:06:55	09:11:36	07:48:08	06:57:14	05:08:26	06:05:09	07:36:18	08:11:47	07:02:23	07:15:07	08:39:47
Daily utilization rate [%]	24.7	9.4	13.6	26.3	44.8	27.2	13.9	8.8	19.1	6.4	9.2

reads were filtered for reasons explained before. The report also shows how many trips were made on a given day. For example, June 17 had only 9 trips due to bad weather (wind gusts prevented the workers from working outside). On April 7, however, significantly more trips were made due to the arrival of windows that had to be moved to the working floors. The average number of trips per day was 50.

In addition to plain numbers, the daily utilization rate of hoisting equipment can be calculated automatically. The following formula was used:

$$\text{Utilization rate} = (\text{Number of trips made} \times \text{Travel time} + \text{Load/unload time}) / \text{Work duration}.$$

The loading and unloading time for each trip of the buck hoist was assumed to be 30 s. Table 2 shows the daily utilization rate of the hoist three. The utilization rate of this lift was measured to be between 6.4 and 44.8%. The 44.8% relate to a day when material was delivered. The idle may vary if the assumption of load and unload time is changed. However, the principle of utilizing the gathered data set for the analysis of the utilization rate of the hoisting equipment remains the same. Such data, in fact, could be used in simulation and project planning to avoid scenarios like workers waiting for buck hoists, as explained before. Means could be applied to maximize the use and minimize of idle time of the hoists as well as other resources.

The study of the gathered data sets is important because they may allow a project manager to visualize the mobility (movement) of resources and also enable them to see specific trends. Allocation of project critical resources can then be prioritized according to the need on certain days and trades. A need for such thoughts exists, for example, when mobilization of resources is necessary because the buck hoist is down for weekly operation and maintenance. Fig. 12 shows two multiple hour delays that the hoist four was out of order. It would not have been efficient for a shipment of windows to arrive during that time of day, or else it could have caused a delay of movement of resources. Such data can thus be used in risk mitigation strategies.

4.4.3. Accounting for personnel and trades, and for site safety and security

The RFID system allowed materials and resources tracking in real time. There was an array of benefits that came with knowing the approximate location of resources at all times on a job site. Other important applications tied into automated time-clocks included site security and emergency response. For instance, if some accident or emergency occurred in a work site, an automated accounting system can easily track the (rough) location (e.g., floor level) of workers. Appropriate actions can be taken to assure their safety and security. Table 3 shows an example of automated timestamp sheet that can be utilized in a construction site. Check-in and check-out time can be automatically tracked and work duration for each worker can be determined. Table 3 shows that a worker, John Henry (name anonymized), showed up at the site 2 h later than other workers (“had a doctor’s appointment”). Such activities can be tracked not only for liability, payroll, and other meaningful purpose but also for confirming site safety for all workers present at a work site and at any time. In the end, the worker stayed longer, with two other teammates. This may have been also the reason why “John” arrived late, to level the available resources for the work task.

The movement of workers on different floors is shown in Fig. 13. The typical work day started at 7:30 A.M. with a stretch and flex exercise, followed by a safety and work plan meeting. These were conducted outside the hoist or work area. The first RFID tag readings thus occurred at around 8:10 A.M. at the platform of the hoist. The workers then proceeded to floor 36, where all tools were stored, before finally starting to work. Movements of workers by trade can be seen in Fig. 13, before leaving the work site they returned to the tool box area to return all tools. The trades left the job site around 4 P.M. Fig. 13 also shows that not all workers used the buck hoist. Such was the case of a journeyman, who started his day on floor 36, where the tools were located. Other means of transportation were accessible to the workers, including the service elevator and stair cases.

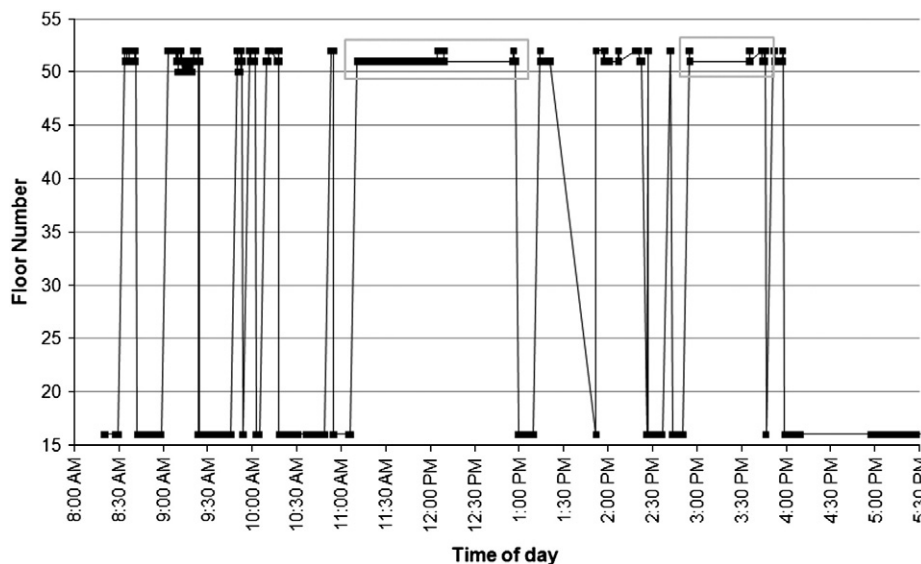


Fig. 12. Temporary down-times of buck hoist #4 in one day.

Table 3

Check-in and -out time for workers (anonymized names).

Last name	First name	Check-in [hh:mm]	Check-out [hh:mm]	Duration [hh:mm]	Last location [Floor level]
Yanz	Miguel	08:19	15:05	06:45	16
Kassin	Tom	08:31	15:05	06:33	16
Davis Jr	Huey	08:19	16:09	07:49	16
Wiseman	Jeff	08:19	14:39	06:20	16
Henry	John	10:23	16:14	05:51	16
Santiago	Mernendez	08:19	16:09	07:49	16

The implementation of a RFID based tracking system thus relies on how well access and entry gates are monitored by technology.

Another important application of the designed RFID system is to allow access to the job site to only those with the authorized permission. Research that even involves drug testing has been done in this domain before, and is now widely applied on many jobs. The developed system, however, is mobile (wireless) and if located accordingly on the job site, for example at the entrance to a confined space, can track any unauthorized entry of workers into such sections of the job site that are either restricted or off limits. Having the ability to track and know where a worker is located is essential in the event of an emergency or evacuation. If an evacuation is ordered, the RFID system can show if and where there is anybody left behind. In case of a true emergency, a response team could quickly locate the last location of the worker(s), greatly reducing the search time or at least defining the start point of a search.

4.4.4. Basic work utilization by trade based on time spent on jobsite

The analysis of the check-in and check-out time of the glazer crew was performed for a week as shown in Table 4. The following assumptions were made to come up a basic work productivity analysis method: The daily work started at 08:15 A.M. and ended at 04:45 P.M. in the afternoon; all crew members were supposed to be present; the initial and final 15 min of a normal work day were not recorded as workers check-in, get their tools, clean-up, return tools, pack-up and check-out; a one-hour overtime was performed during the observation time; at least five glazers build a functioning crew; only one crew can work on a given floor at a time; break and lunch time are, but travel time, personal time, etc. are not considered in the analysis. This data can then be used hand-in-hand with the physical output (daily, weekly, monthly etc.) to perform the productivity analysis, since it provides the time spent on the work.

Table 4

Check-in/check-out time analysis of glazers.

Glazer [#]	Discrepancy [hh:mm:ss]				
	Jun 21	Jun 22	Jun 23	Jun 24	Jun 25
5	01:44:04				
6	01:56:42				
7	06:42:36				
9	00:40:17	00:44:26	00:41:04	00:42:29	00:40:28
10	02:09:28		05:37:51	00:38:11	00:40:26
12			00:43:50	00:42:26	00:40:57
18	02:38:28	02:20:21	00:36:14	00:36:09	00:39:00
19	00:40:21	00:43:34		00:38:00	00:40:49
Average	02:21:42	01:16:07	01:54:45	00:39:27	00:40:20
Average per worker per day					01:22:28
Percentage of total considered time [%]					16.2

The following analysis was performed on a weeklong data set that was randomly picked from the project. The RFID tags of a subcontracted crew of glazers were selected to limit the demonstration the analysis of the worker utilization. Under the assumptions previously made, information was generated from the RFID data set. The result is illustrated in Table 4 in form of missed work times. The work time lost per day per worker was around 1 h 22 min. This time amounts to 16.2% of the expected work time.

However, these values did not include work performed on other floors, e.g. preparatory work. Removing data for glazer #7 on June 21 and glazer #10 on June 23 (both worked on a lower floor level assisting another crew; crew were thus spatially separated from each other) results automatically in an average lost time per worker per day of about 11.5% (see Table 5).

Individual analysis was performed for two of the glazers (#9 and #18). The results in Table 6 show an average of 42 min for glazer #9 and 1 h 22 min for glazer #18 that was lost on a daily basis, which are 8.2% and 16.1% of the total considered work time, respectively. Reasoning to why the time was lost can only happen if other data is linked, e.g. weather data. This was not the case. However, the principle of how the data can be used has been shown.

4.4.5. Travel time analysis of a construction manager

What can be done for buck hoists, workers or trades, can also be done for personnel associated to management. An analysis of the travel time of a construction manager was performed. Fig. 14 and Table 8 indicate that the manager was present on the 16th floor at

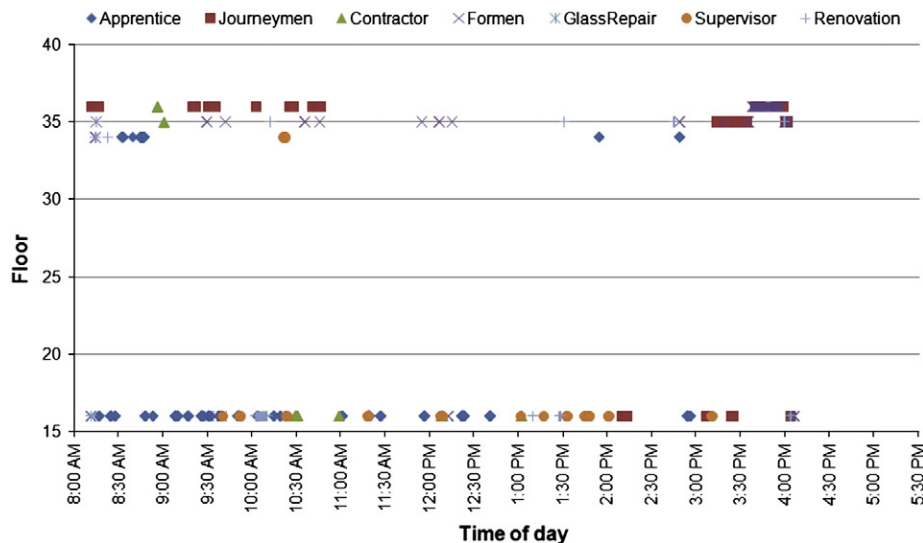


Fig. 13. Graph of worker tags picked up by the readers on their designated floors on June 25, 2010.

Table 5
Check-in/check-out time analysis of glazers.

Glazer [#]	Discrepancy [hh:mm:ss]				
	Jun 21	Jun 22	Jun 23	Jun 24	Jun 25
5	01:44:04				
6	01:56:42				
7					
9	00:40:17	00:44:26	00:41:04	00:42:29	00:40:28
10	02:09:28			00:38:11	00:40:26
12			00:43:50	00:42:26	00:40:57
18	02:38:28	02:20:21	00:36:14	00:36:09	00:39:00
19	00:40:21	00:43:34		00:38:00	00:40:49
Average	01:38:13	01:16:07	00:40:23	00:39:27	00:40:20
Average per worker per day [hh:mm:ss]					00:58:54
Percentage of total considered time [%]					11.55

08:58 A.M. The first trip the manager made was to floor 33 at 10:53 A.M.. Tables 7 and 8 indicate that it took 2 min 40 s to go up to floor 35. The manager then took the stairs to the next lower level (either to not interfere with material hoist, or because the wait time was too long or inconvenient), which means the manager spend a total of 19 min 27 s on both floors (35 and 34). At a later time of the day (01:37 P.M.), the manager made one more trip with the buck hoist to floor 35. The manager spends about 32 min on the floor to resolve an unknown issue. The second trip was completed when traveling back to floor 16 (where the site offices were located) at 14:09 P.M.. Such data can illustrate the amount of supervision or inspection (time) that is given on a daily basis. Though the manager was present on site for 7 h and 27 min on June 24, the manager made just two relatively short trips to the upper floors. This means much more work was going on in the offices located on the 16th floor.

4.4.6. Travel time trend analysis

The gathered data can also be analyzed for trends. This was done based on time taken for the buck hoist and buck operator to travel from the lower floor to upper floors, upper floors to lower floor as well as among the upper floors moving up and down. Fig. 15 shows that the travel time from the upper floors to lower floor is less than the travel time from lower floor to the upper floors. The reason may be because of gravitational factors. Some days seemed to have had greater travel time due to various factors (e.g. material delivery, load). Fig. 15 also shows the trend of travel time reducing over the project duration. This happened because of the fact that work was initially started from higher floors and gradually the crew moved down after completion of work at higher floors. As a result, the readers were also moved down.

As shown in Fig. 15, the curves for the travel time of lower part of the upper floors to the higher part of the upper floors overlaps with the travel time from the higher upper floors to lower upper floors. Since other constant times like opening and closing the door, mobilization time etc. become more significant in this case. Because of its short span, travel time becomes less dominant. Fig. 16 shows similar analysis done with the buck hoist. Analysis done for travel time of buck hoist operator in Fig. 15 and that for buck hoist in Fig. 16 gave similar results, which was as expected.

Table 6
Check-in/check-out time analysis of glazers present throughout the week.

Glazer [#]	Discrepancy [hh:mm:ss]					Avg. per day [hh:mm:ss]	Percentage of time [%]
	Jun 21	Jun 22	Jun 23	Jun 24	Jun 25		
9	00:40:17	00:44:26	00:41:04	00:42:29	00:40:28	00:41:45	8.19
18	02:38:28	02:20:21	00:36:14	00:36:09	00:39:00	01:22:02	16.09

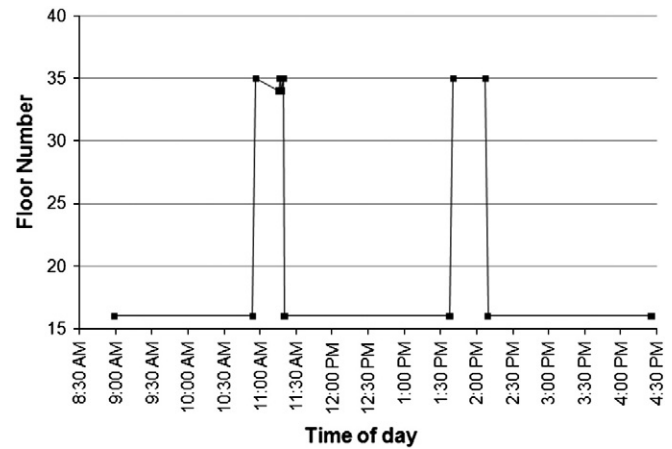


Fig. 14. Activity of the construction manager on June 24.

Table 7
Travel details for the construction manager on June 24.

Start time (hh:mm:ss)	End time (hh:mm:ss)	Floors traveled	Travel time (hh:mm:ss)
10:53:46	10:56:26	19	00:02:40
10:56:26	11:15:53	1	00:19:27
11:16:45	11:17:56	1	00:01:11
11:19:46	11:20:23	19	00:00:37
13:37:48	13:40:38	19	00:02:50
14:07:25	14:09:27	19	00:02:02

5. Cost vs. benefit of automated field mobility and status monitoring passive RFID system

There are many sophisticated systems that can save time and reduce labor at a construction site. Many project site monitoring systems, whether manual or (semi-) automated, cannot be implemented because of economic reasons. A cost–benefit comparison can help in determining whether a system is efficient or not. The cost includes the necessary investment in successful implementation of an automated field mobility and status monitoring system, along with maintenance and operation throughout the considered project period. The benefit includes the money saved or productivity gained because of the implementation of the system. If the benefits surpass the costs the system is considered to be feasible. The gain from investment minus the cost of investment divided by the cost of investment is called the minimum return-on-investment (ROI), which gives an idea of the scale of the benefits that a customer will receive once using it.

A cost–benefit analysis was done to determine the economic viability of the passive RFID system. The estimated costs include the purchase of RFID antennas, computers, database and networking infrastructure, cables, and tags to track a reasonable amount of personnel, equipment, and material. A fixed fee for installation and maintenance was assumed per month. The benefits results from cost savings, assuming that seven workers (\$20 per hour) are 3 h late, just once a month. Since many subcontractors are hired on lump sum contracts, their time might not be valued in a way that it would result in a saving to

Table 8

Work and travel statistics of the construction manager on June 24.

	[hh:mm:ss]
Start time	08:58:48
End time	16:25:33
Total work time	07:26:45
Number of trips	6

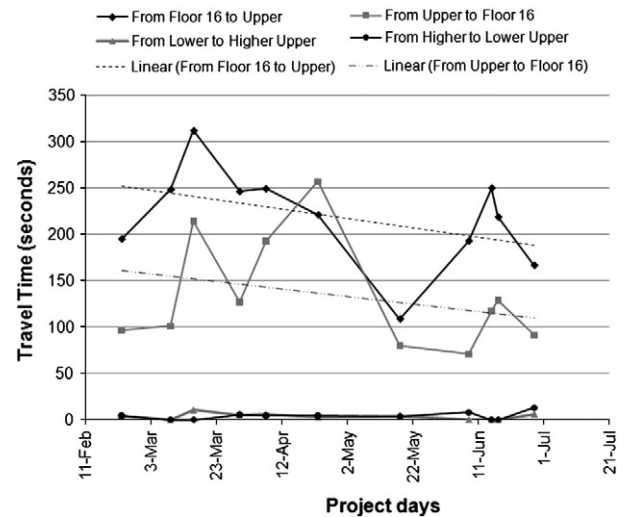
the general contractor or construction manager which might install the system, thus more important savings are expected from savings in reduced management time. A construction engineer, for example, may earn \$80 per hour, and may spend on average about 1 h each work day on checking manual lists that subcontractors turned in. This task alone could be valued at \$38,400. Additionally, the automated system could also save other management time, e.g. material tracking, assist in legal claims, site documentation, automated billing, task scheduling, security and safety enablers, and assisted work sampling studies, that each carry other benefits and result in value.

The rough analysis as shown here, has a minimum return on investment of 2.2 on its first use. The ROI increases if the same technology is used on a future project again. Hence, automated tracking of construction resources leveraging passive RFID technology can be considered feasible, even economically.

<i>Estimated costs</i>	
6 wireless RFID readers (\$1000 each)	\$6000
3 netbook computers (\$300 each)	\$900
1 database (\$5000) and wireless networking infrastructure (\$200)	\$5200
7000 tags (\$0.10 each)	\$700
Installation and Maintenance (\$2000 per month, 2 years)	\$24,000
Total	\$31,800
<i>Estimated benefits</i>	
Automated billing of one subcontractor's work crew	
If 7 workers are 3 h late, once a month, \$20/h per worker, 2 years	\$62,400
Reduced management time, 24 months project, 1 h per work day, \$80/h	\$38,400
Total	\$102,400
Minimum Return on Investment (ROI) on first project	2.2

6. Limitations and barriers

Certainly, technological limitations must be carefully evaluated before implementing. These may not only related to the RFID technology alone, e.g. read range, multipath, but mainly to software, data

**Fig. 16.** Travel time analysis of hoist #3 throughout the project.

storage, and data processing tools that yet has to be defined and created for reliable operation in harshest construction applications.

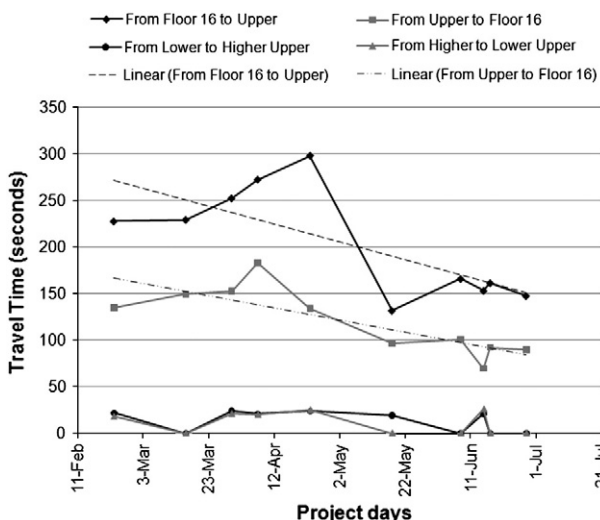
The acceptance of workers was also crucial in the success of implementing technology in the field. As most workers welcomed the technology, some requested extra explanation (that this technology does not track their location) before they were willing to use the technology. Some subjects were not cooperative at first, and there have been a few cases of equipment sabotage (e.g., pulling the power plug). However, once training and cooperation were enacted, there was full support and cooperation from the volunteers. As an additional finding of this study, contracts may need to be set up in a way that understanding among general contractor and subcontractors exist whether a particular technology is used on a project, and such a decision should be better made at the front-end of a project.

7. Conclusions and recommendations

Active RFID technology has been proven in past research to benefit the construction industry in various outdoor environments. This research demonstrated results for passive RFID technology in an indoor construction environment where lifting capacity and presence of trades is crucial for project success. This paper explained the design and implementation of a wireless passive RFID technology system that was successfully implemented in a high-rise renovation project in order to record important project data.

A lengthy experiment over several months was conducted to determine the feasibility of implementing the technology in practice. Despite the vast height of the structure and the presence of concrete and metallic elements, the technology worked fine. Furthermore, signal strength data showed that passive RFID tags can be read by readers whether placed on the exterior or interior of a hard hat. It is suggested for future use that passive tags should be embedded in the interior of a hard hat to reduce tampering by workers.

In the past, analysis of efficiency of construction resources (personnel, equipment, material) has required substantial amounts of human labor hours to collect and analyze data, that eventually was prone to human error. The implementation of RFID can greatly improve the accuracy of such analysis while automating the task. The use of a database also showed to benefit staff members' time for preparing manual documents. The analysis conducted in this research showed that the installed buck hoist system, while less time efficient in material transfer than the commonly used tower crane in high-rise settings, is a sufficient means of transportation for larger work crews of fifty or more people. Efficiency studies, such as buck hoist availability, had become feasible utilizing the designed passive RFID system.

**Fig. 15.** Travel time analysis of buck hoist operator.

Data presented in this paper demonstrates that lift availability depends on work tasks scheduled during the day.

Future research should focus on fully automating the data transfer process from the jobsite to a database for site operation and scheduling analysis. Other future studies should focus on the impact of technology on humans, for example, understanding workers' motivation to participate in RFID studies and in particular when RFID tags are deployed on their hardhats. Overall studies may be performed on understanding how all project stakeholders (including workers) can benefit from using advanced sensing technology in the field. Technical studies should be conducted to how passive RFID technology and in particular the read distance of passive RFID tags can be improved, and how data can be integrated in building information models (BIM). In summary, this paper was an initial research project towards the feasibility of leveraging a passive RFID system for project field mobility and status monitoring that comes at affordable cost in a high-rise renovation setting.

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